Wednesday 17th December 2025.

Lagrange Multipliers.

This is a method used for finding maximum and minimum of constrained function. That is, finding maximum and minimum of $f(x,y,z)=0$ subject to $\phi(x,y,z)=0$.

This consists of forming an auxilliary function $G(x,y,\lambda) = f+\lambda\phi$ or $\lambda f+\phi=0$. The λ is langrange multiplier.

To find the relative extrema of the function $f(x,y)$ subject to the constrained $\phi(x,y)=0$ (assuming this extrema value exists.

1. Form an auxillary equation $G(x,y,\lambda) = f(x,y) + \lambda\phi(x,y)$ called the Lagrangian function. (the variable λ is called the Lagrange multiplier.

2. Solve the system that consists of equation i.e

$G_x = 0$; $G_y = 0$, $G\lambda = 0$ for all x, y and λ.

3. The solution found in step 2 are candidates of the extrema of 'f'.

Example 1.

Use the method of langrange multiplier to find the relative minimum of the function $f(x,y) = 2x^2 + y^2$ subject to the constrained $x + y = 1$

Soln

The constrained equation $x+y=1$ can be written in the form $\phi(x,y) = x+y-1 = 0$

step 2: Form the lagragian function $G(x,y,\lambda) = f(x,y) + \lambda\phi(x,y)$

$$= 2x^2 + y^2 + \lambda(x+y-1)$$

$$2x^2 + y^2 + \lambda x + \lambda y - \lambda$$

$$G_x = 4x + \lambda = 0 \quad \text{--- (i)}$$

$$G_y = 2y + \lambda = 0 \quad \text{--- (ii)}$$

$$G\lambda = x + y - 1 = 0 \quad \text{--- (iii)}$$

Solving equ (i) and (ii) for x and y in terms of λ to give:

$$4x + \lambda = 0$$

$$x = \frac{-\lambda}{4} \quad \text{--- (iv)}$$

$$2y + \lambda = 0$$

$$y = \frac{-\lambda}{2} \quad \text{--- (v)}$$

From equ (iii)

$$\frac{-\lambda}{4} - \frac{\lambda}{2} - \frac{1}{1} = 0$$

$$\frac{-\lambda - 2\lambda - 4}{4} = 0$$

$$\frac{-3\lambda - 4}{4} = 0$$

$$-3\lambda - 4 = 0$$

$$-3\lambda = 4$$

$$\frac{\cancel{-3}\lambda}{-3} = \frac{4}{-3} \quad \lambda = -4/3$$

From equ (iv), $x = -\frac{1}{4}\left(-\frac{4}{3}\right) = \frac{1}{3}$

$y = -\frac{1}{2}\left(\frac{-4}{3}\right) = 2/3$.

Therefore $x = 1/3$ and $y = 2/3$

Thus, $f_{min}\left(\frac{1}{3}, \frac{2}{3}\right) = 2\left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2$

$= \frac{2}{9} + \frac{4}{9} = \frac{6}{9} = \frac{2}{3}$ //.

Example 2.

Find the minimum value of $x^2 + y^2 + z^2$ when $x + y + z = 3a$.

~~[illegible]~~

Soln

$F(x, y, z) = x^2 + y^2 + z^2$

$\phi(x, y, z) = x + y + z - 3a = 0$

Form the auxillary equation

$G(x, y, z, \lambda) = f(x, y, z) + \lambda\phi(x, y, z)$

$= x^2 + y^2 + z^2 + \lambda(x + y + z - 3a)$

$= x^2 + y^2 + z^2 + \lambda x + \lambda y + \lambda z - 3a\lambda$

$G_x = 2x + \lambda = 0$

$G_y = 2y + \lambda = 0$

$G_z = 2z + \lambda = 0$

$G_\lambda = x + y + z - 3a = 0$

At maximum/minimum point

$x = -\lambda/2 \quad y = -\lambda/2 \quad z = -\lambda/2$

$G_x = G_y = G_z = G_\lambda = 0$

$\lambda = -2x \quad \lambda = -2y \quad \lambda = -2\lambda$

$\Rightarrow -2x = -2y = -2z$

But $x + y + z - 3a = 0$

$x + x + x - 3a = 0$

$3x - 3a = 0$

$\frac{\cancel{3}x}{\cancel{3}} = \frac{\cancel{3}a}{\cancel{3}} \quad x = a$.

$\lambda = -2a$.

Hence $y = a$, $z = a$

$(x, y, z) = (a, a, a)$

The required minimum value of $x^2 + y^2 + z^2$ is $a^2 + a^2 + a^2 = 3a^2$.

Exercise

Find the minimum value of $x^2 + y^2 + z^2$ subject to the condition $xyz = a^3$

2. Find the maximum and minimum values of the function $f(x, y) = 3x + 4y$ subject to the constrained $x^2 + y^2 = 1$

3^{4n6} $48 + \frac{1}{x}$

(3) Find the stationary point of the function $f(x,y,z) = x^2+2y^2+z$ subject to the constrained $\phi(x,z) = x^2 - z^2 - 2 = 0$

(4) Find the ~~maxima and minima of $f(x,y,z) = xy^2z^3$ subject to the condition $x+y+z=6$, $x>0$~~, $y>0, z>0$.

5th January, 2026.

Differential

Let the function $y = f(x)$ with $\Delta y = f'(x)\Delta x + \alpha\Delta x$, then the function given $dy = f'(x_0)\Delta x$

Example.

Find the differential of the function $y = 1/x$ at the point $x = 2$

Solution

$x_0 = 2$.

$f'x = \frac{-1}{x^2}$ $f'(2) = -1/4$

$\Rightarrow -\frac{1}{4}\Delta x$.

Example 2:

$y = x^2$ at point $x_0 = 20$ and $\Delta x = 0.1$. Find dy and Δy.

Soln

$\Delta y = f(x_0 + \Delta x) - f(x_0)$

$= (x_0 + \Delta x)^2 - x_0^2$

$= (20 + 0.1)^2 - 20^2$

$= 4.01$.

$dy = 2x_0 \Delta x$

$= 40 \times 0.1$

$= 4.0$

Example 3:

Find the differential of $y = x^3 + \sqrt{x}$

$f'(x) = 3x^2 + \frac{1}{2\sqrt{x}}$

$dy = \left(3x^2 + \frac{1}{2\sqrt{x}}\right)\Delta x$.

4) Find the differential of $y = x^2\sin(x-2)$ at $x = 2$

(ii) $y = \sqrt[3]{x}$

Approximations.

Suppose the function $y = f(x)$

$\Delta y = f'(x_0)\Delta x + \alpha \Delta x$

($\alpha \to 0$ and $\Delta x \to 0$).

$\Delta y - f'(x_0)\Delta x = \alpha \Delta x$

$\Rightarrow \Delta y \simeq f'(x_0)\Delta x$

$\Rightarrow f(x_0 + \Delta x) \simeq f(x_0) + f'(x_0)\Delta x$

Example

Find the approximate value of $f(x) = \sin 46$

Soln

$f(x) = \sin x \quad f'(x) = \cos x$

$\sin 46 = \sin(45^\circ + 1) \simeq (\sin \frac{\pi}{4} + \sin \pi/180) \cos \pi/4$

$\sin 46^\circ \simeq \sqrt{2}/2 + \pi/180 \sqrt{2}/2$

$= 0.7071 + (0.7071)(0.0175)$

$= 0.7191$

Example.

$\sqrt[3]{x + \Delta x} \simeq \sqrt[3]{x} + (\sqrt[3]{x})' \Delta x$

evaluate $\sqrt[3]{1.02}$

Find the approximate value of $y = \sqrt[3]{x}$

$\sqrt[3]{x + \Delta x} \simeq \sqrt[3]{x} + (\sqrt[3]{x})' \Delta x$

$x = 1, \Delta x = 0.02 \simeq \sqrt[3]{1} + \frac{1}{\sqrt[3]{1}}(0.02) \simeq 1.0066$

Example

$\simeq \sin 29^\circ$

$\sin(x + \Delta x) = \sin x + (\sin x)' \Delta x$

$\sin(30 - 1) \simeq \sin \frac{\pi}{6} + \cos \frac{\pi}{6}\left(\frac{-\pi}{180}\right)$

$\simeq 1/2 - \frac{\sqrt{3}}{2}\frac{\pi}{180}$

$\simeq 1/2\left(1 - \frac{\pi\sqrt{3}}{180}\right)$

The linearization of $f(x,y)$ where x is differentiable $L(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x - x_0) + f_y(x_0,y_0)(y - y_0)$

the approximation $L(x,y) \simeq f(x,y)$ is the linear approximation

Example

Find the Linearization of

$f(x,y) = x^2 - xy + \frac{1}{2}y^2 + 3$ at the Point $(x_0, y_0) = (3, 2)$

Solv:

$f(x_0, y_0) = 3^2 - 3(2) + \frac{1}{2}(4) + 3$

$= 9 - 6 + 2 + 3 = 8.$

$f_x(x_0, y_0) = 2x - y|_{3,2} = 4$

$f_y(x_0, y_0) = -x + y|_{3,2} = -1$

$L(x,y) \simeq 8 + 4(x-3) - 1(y-2)$

$8 + 4x - 12 - y + 2 \simeq 4x - y - 2$

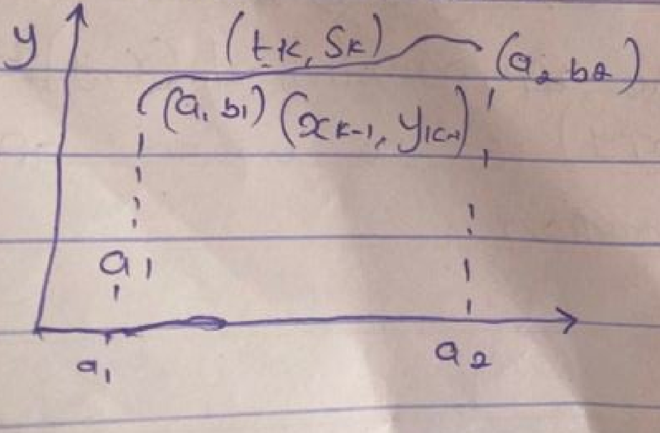
Exercise.

$L(x,y)$ or $L(x,y,z)$

(i) $f(x,y) = x^2 + y^2 + 1$ at $(1,1)$

(ii) $f(x,y,z) = xy + yz + xz$ at $(1,1,1)$

LINE INTEGRAL



Let C be the curve connecting $a_1 b_1$ and $a_2 b_2$ in the xy plane, the limit of the sum

$$\sum_{k=1}^{n} \{ P(t_k, s_k)\Delta x_k + Q(t_k, s_k)\Delta y_k \} \text{ as } \Delta x_k, \Delta y_k \to 0$$

$\Delta x_k = x_k - x_{k-1}$

$\Delta y_k = y_k - y_{k-1}$

The limit is denoted by

$$\int_C P(x,y)\,dx + Q(x,y)\,dy$$

OR

$$\int_{(a_1,b_1)}^{(a_2,b_2)} P\,dx + Q\,dy.$$

If C is given in parametric form $x = \phi(t)$: $y = \psi(t)$

the Integral becomes :

$$\int P(\phi(t), \psi(t))\,\phi'(t)\,dx + Q(\phi(t), \varphi(t))\,\psi(t)\,dt$$

Evaluate

$I = \int_c (x^2 + y)\,dx + (x - y^2)\,dy$

from $A(0,2)$ to $B(3,5)$

$y = 2 + x$

$I = \int P\,dx + Q\,dy$

$dy = dx.$

$= \int_0^3 (x^2+2+x)dx + [x-(2+x)^2 dy]$

$= \int_0^3 (x^2+2+x)dx + [x-(4+4x+x^2)dy]$

$= \int_0^3 (x^2+2+x)dx + [x-4-4x-x^2]dy$

$= \int_0^3 (x^2+2+x)dx + [-x^2-3x-4]dx$

$\int_0^3 [x^2+x+2]dx - [x^2+3x+4]dx$

$= \int_0^3 (-2x-2)dx$

$= \int_0^3 -(2x+2)dx = -[x^2+2x]_0^3$

$-[3^2+2(3)] = -[9+6] = -15.$

Example 2.

Evaluate

$\int_{(0,1)}^{(1,2)} (x^2-y)dx + (y^2+x)dx$

along (a) Straight line from $(0,1)$ to $(1,2)$

(b) Straight line from $(0,1)$ to $(1,1)$ and $(1,1)$ to $(1,2)$

(c) $x=t,\ y=t^2+1$

Soln

(a) $y = x+1 \quad dy = dx$

$I = \int_0^1 [x^2-(x+1)]dx + [(x+1)^2+x]dx$

$= \int_0^1 [x^2-x-1+x^2+2x+1+x]dx$

$\int_0^1 [2x^2+2x]dx$

$\frac{2x^3}{3} + \frac{2x^2}{2}\Big|_0^1$

$\frac{2(1)^3}{3} + \frac{2(1)^2}{2}$

$\frac{2}{3} + \frac{2}{2} \Rightarrow \frac{4+6}{6} \Rightarrow \frac{10}{6}$

$\Rightarrow 5/3\ ..$

(b) $dy = 0$

$I_1 = \int_0^1 (x^2-1)dx + (1+x)\cdot 0$

$= \int_0^1 (x^2-1)dx = \frac{x^3}{3} - x\Big|_0^1 = -\frac{2}{3}$

$(1,1)$ to $(1,2)$ $x=1$ $dx=0$

$I_2 = \int_1^2 ((1-y)0 + y^2+1)dy$

$= \int_1^2 (y^2+1)dy = \frac{y^3}{3} + y\Big|_1^2 = \frac{10}{3}$

$I = I_1 + I_2 = \frac{-2}{3} + \frac{10}{3} = \frac{8}{3}.$

(c) $x = t \qquad t=0 \quad t=1$

$I = \int_0^1 [t^2-[t^2+1]]dt + [(t^2+1)^2+t]\,2t\,dt$

$= \int_0^1 [t^2-t^2-1+[t^4+2t^2+1+t]2t]dt$

$= \int_0^1 [-1+2t^5+4t^3+2t^2+2t]dt$

$$= -t + \frac{\cancel{2}^{1} t^6}{\cancel{6}\,3} + \frac{\cancel{4} t^4}{\cancel{4}} + \frac{2t^3}{3} + \frac{\cancel{2} t^2}{\cancel{2}} \Big|_0^1$$

$$= -t + \frac{t^6}{3} + t^4 + \frac{2t^3}{3} + t^2 \Big|_0^1$$

$$= (-1) + \frac{(1)^6}{3} + (1)^4 + \frac{2(1)^3}{3} + (1)^2$$

$$\Rightarrow \frac{-1}{1} + \frac{1}{3} + \frac{1}{1} + \frac{2}{3} + \frac{1}{1}$$

$$\Rightarrow \frac{-\cancel{3} + 1 + \cancel{3} + 2 + 3}{3} \Rightarrow \frac{6}{3} = 2$$